A quick introduction to knime

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A quick Introduction

Open for Innovation

KNIME

knime

- **KNIME Analytics Platform** is free and open-source data analytics, reporting, and integration platform.
- It allows users to build workflows for machine learning, data mining, ETL (Extract, Transform, Load), and visualization without requiring extensive programming.
- No code / Low Code Visual Programming Tool for Data Analytics and ML
- Provides a drag-n-drop interface for building data workflows, also called data pipelines.
- Execution components are called nodes, which are configurable blocks, designed to perform a function.

How to get knime analytics platform on your system?

Step-by-Step Instructions

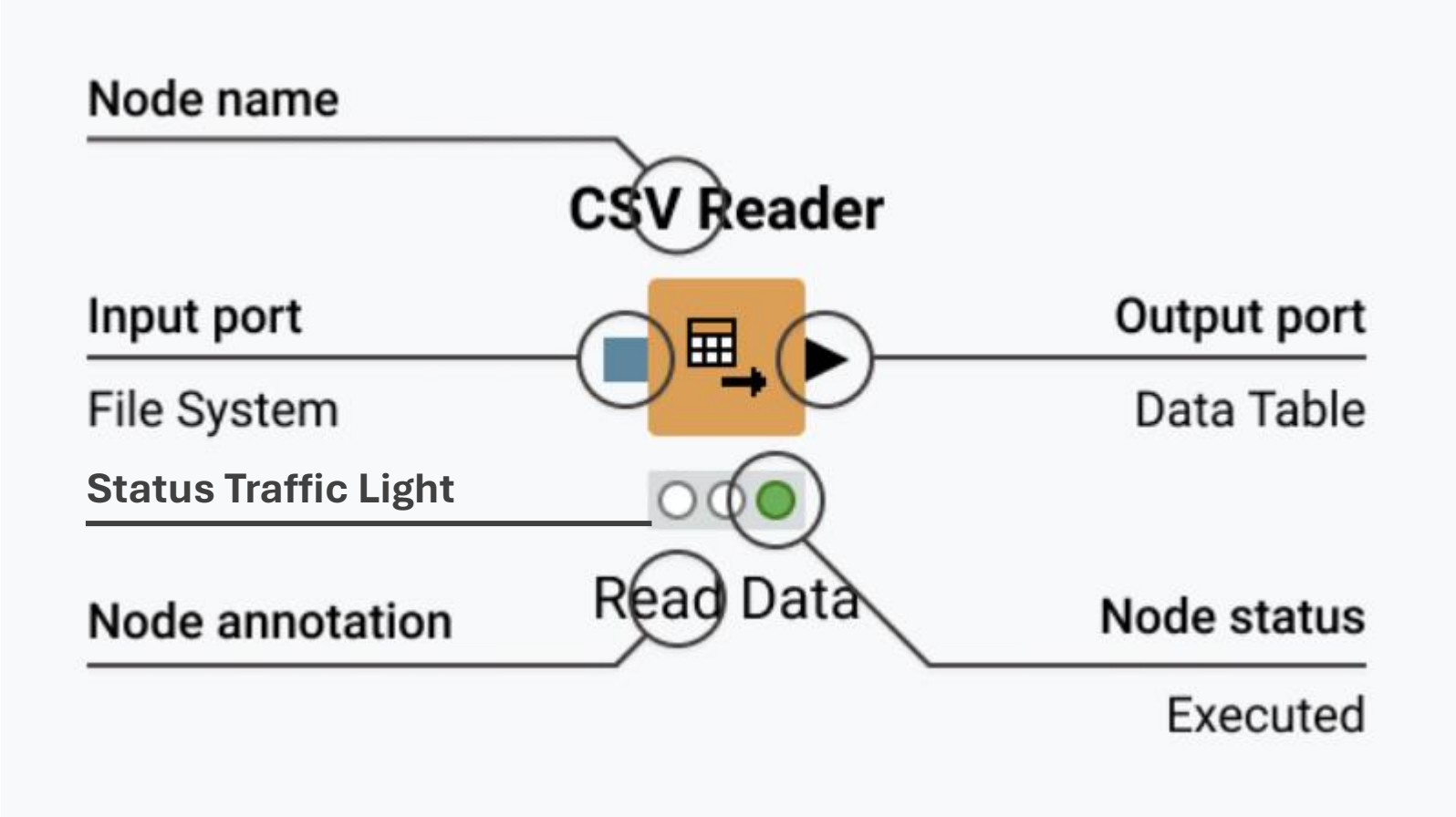
- Go to the Official Website
  - Visit https://www.knime.com/downloads
- Choose Your Operating System
  - Select your platform: Windows, macOS, or Linux.
- Download the Installer
- Click the appropriate download link for your OS. For Windows, it’s typically a .exe file.
- Run the Installer (Run as Administrator)
- On Windows: Double-click the .exe file and follow the setup wizard.
- Launch KNIME
- After installation, open KNIME and start building workflows!

The workflow (data pipeline)

- A workflow defines the data flow pipeline for the application.
- It is made up of several connected nodes.
  - Select the desired node, drag into the workspace and connect them to form a data pipeline. The pipeline indicates the flow of data from one node to the other.
- Node is the execution block that performs a function.
- Nodes can be configured
- Nodes will have connection ports (input ports / output ports) that connect and communicate with other nodes to receive or send data

The Node, code block

The Execution Context, a function block.



Node Status

The **Status Traffic Light** indicates the current status of the node.

Not configured
The node is waiting for configuration or incoming data.

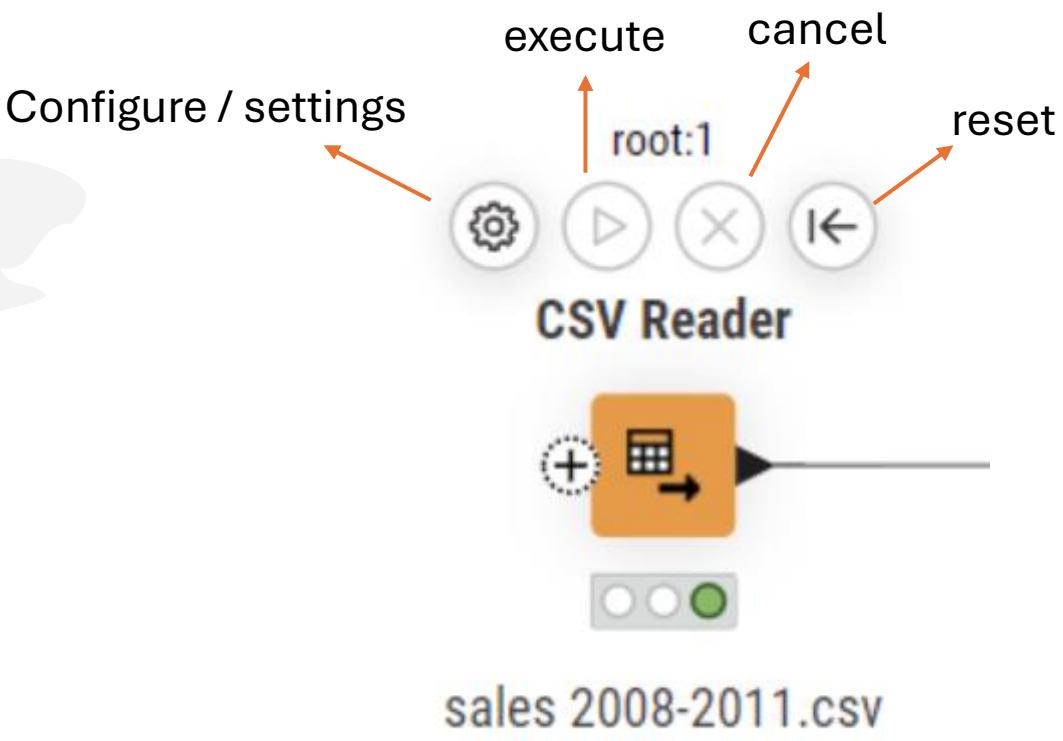
Configured
The node has been configured correctly, and can be executed.

Executed
The node has been successfully executed. Results may be viewed and used in downstream nodes.

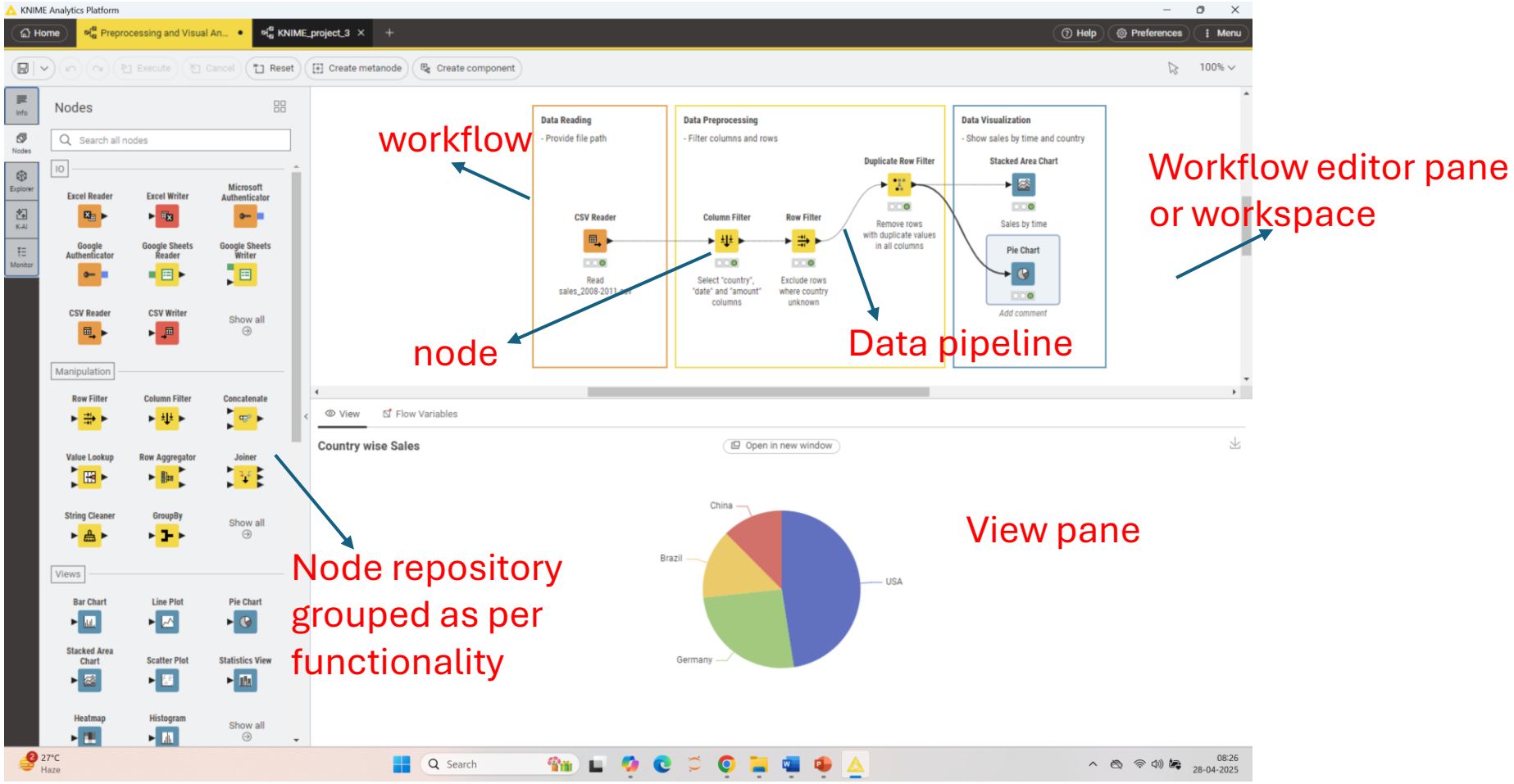
Error
The node has encountered an error during execution.

Node Configuration

Hover mouse on the node to expose options



knime in Action: The dashboard



A small tour of knime

End of Session

Thank You